Research summary | Master's thesis

Coupled Deblurring and 3D Reconstruction from Image Sequences

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The microscopic camera CAM-M, mounted on the lunar rover Rashid-1 for close up image acquisition of the lunar regolith surface, has no refocus mechanism, resulting in a shallow depth of field that produce image with defocus blur and in turn reduced number of feature correspondences. The 3D reconstruction of a sequence of those overlapping defocused images can provide how much each pixel have deviated from the plane of focus which can be used to estimate blur and restore images sharpness. Images are deblurred using this geometry-derived, spatially varying blur model in Wiener filter and Richardson-Lucy deconvolution producing a 3D reconstruction with a significantly increased areal point density and thus a much higher spatial resolution than the initial 3D reconstruction. The developed method was successfully applied to CAM-M image sequences taken during a lunar rover test drive at the MBRSC.

Context and problem:

The microscopic camera CAM-M on the Rashid-1 lunar rover is designed to capture images of the undisturbed regolith at 20-27 µm per pixel, obtaining grain-scale roughness for thermal and geological models of the lunar surface [1]. CAM-M has no focus or aperture adjustment mechanism, its depth of field is 22 mm and its optical axis is tilted 40° from vertical. These features lead to uneven terrains and rover pitch and roll pushing parts of every frame out of focus. The blur therefore is not uniform and varies spatially across the image. Depth-from-focus and depth-from-defocus methods require focus stacks or several apertures settings, respectively, both of which the microscopic camera CAM-M by design cannot provide.
Research question: can spatially varying defocus blur be estimated and deblurred using only overlapping image sequences from monocular camera with a fixed depth of field (focus) and aperture?

Approach:

The key concept is to use structure from motion (SfM) and Multi-view stereo (MVS) reconstruction to obtain missing depth information, camera poses and 3D surface points. This information is combined with the known optic properties of the camera, providing a physically grounded blur estimate for each pixel.

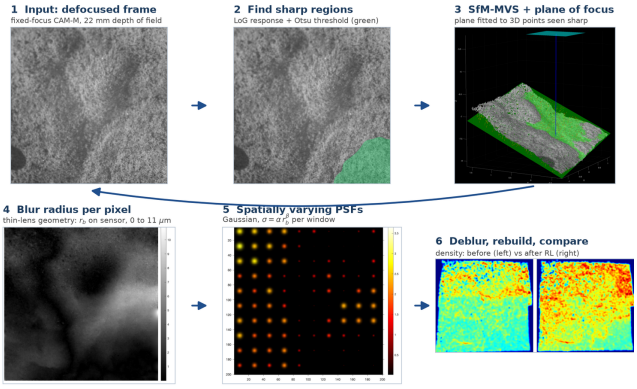


Figure 1. Method overview (Data set 1, image 2). (1) Defocused input image. (2) Sharp region (green) from LoG and Otsu thresholding. (3) SfM-MVS point cloud with fitted plane of focus (green), optical axis (blue) and camera pose (cyan). (4) Per-pixel circle of confusion r_b image array. (5) Sampled local Gaussian PSFs: larger and brighter spots indicate stronger local blur. (6) Point density map before (left) and after (right) Richardson-Lucy deblurring on a common scale.

1. **Reconstruct** the initial 3D point cloud from the grayscale 1024x1024 image sequence using SfM-MVS in Agisoft Metashape Pro, scripted through its Python API.
2. **Estimate the plane of focus (sharpness)**: Sharp, focused regions are detected with a Laplacian of Gaussian and Otsu thresholding. These regions are then matched to projected 3D tie points of the initial point cloud. A plane of focus is fitted to the corresponding 3D points using SVD, however six other fitting methods were implemented as well.
3. **Derive Blur from optics geometry**: For each 3D point, its distance and angle to the optical axis provides the defocus on the object's side of the image formation model. The thin-lens optics equation provides the shift on the image plane side of the model along with circle-of-confusion radius r_b. The metric scale comes from the known working distance, and pixels with no radius values are interpolated to obtain a dense per-pixel r_b image array serving as a map.
4. **Spatially varying deconvolution**: A Gaussian point spread function (PSF) with $\sigma = \alpha(r_b)^{\beta}$ is built for each 8x8 pixel sliding local window of which the inner region is 4x4 pixel with 50% overlap. The image is restored by a windowed approach with both the linear Wiener filter and the nonlinear,

iterative Richardson-Lucy algorithm. For the Wiener filter, the noise-to-signal ratio comes from a median-filter noise estimate with an empirically tuned parameter.

5. **Evaluate on the downstream task**: The restored deblurred images reconstruct the 3D point clouds. The point clouds are then aligned with ICP and normalized foe efficient comparison of the point density maps.

Data: Two image sequences provided by the CAM-M science team's 2023 field tests in the MBRSC Moon Yard using regolith simulants. In each data set, three overlapping images were deblurred.
Implementation: MATLAB for geometric modeling and analysis, method's computations and deconvolution. Python for Metashape API automation, data analysis and extraction, and image preprocessing.

Results:

- Deblurred images gave **denser clouds with fewer data gaps** in both data sets, a significantly **increased areal point density** and thus a much **higher spatial resolution** than the initial 3D reconstruction.
- **Method's performance is dependent on data**: In data set 1, Richardson-Lucy provided the densest point cloud with almost no remaining gaps, while Wiener filter produced a denser overall point cloud but introduced few small new gaps. In data set 2, Wiener filter obtained the densest point cloud and recovered the bead surface structure, but produced new gaps in the upper right corner. RL had fewer gaps but was less dense overall. Across both sets, Wiener filter was the most consistent of the two methods.

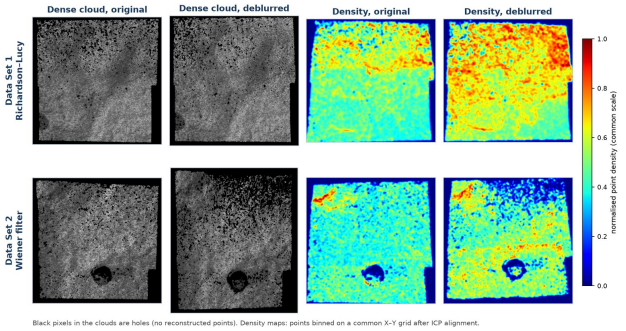


Figure 2. Effect on 3D reconstruction. Top: Data set 1 with Richardson-Lucy (5 iterations). Bottom: Data set 2 with the Wiener filter. Columns 1-2: dense point clouds in the X-Y view. Columns 3-4: normalized point density on a common scale after ICP alignment.

- **Visual quality is not always an indication of providing more useful information**: some settings that produced visibly sharper images produced fewer feature correspondences, so restoration should be evaluated based on the downstream output quality, information gain and tuned for the geometric task.

Limitations:

- The evaluation rests on the comparison between density maps, with no ground-truth surface reconstruction. The parameters were tuned per image to maximise density on the same data.
- Intrinsics were estimated by self-calibration in bundle adjustment and the scale was obtained from the nominal working distance.